STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-FIBRE-PRODUCTS-PRESERVE-PROPERTIES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2944-2955 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2957-3000, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_algebraic-stacks-sheaves.json (pb=1732, stmt 2944-2955, proof 2957-3000).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0040: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3000–3000. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-fibre-products-morphism: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2914–2920. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$, $g : \mathcal{Z} \to \mathcal{Y}$ be faithful 1-morphisms of categories fibred in groupoids over $(Sch/S)_{fppf}$.*

(1) *the functor $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z} \to \mathcal{Y}$ is faithful, and*
(2) *if $\mathcal{X}, \mathcal{Z}$ have equalizers, so does $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$.*

Proof. We think of objects in $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$ as quadruples (U, x, z, α) where $\alpha : f(x) \to g(z)$ is an isomorphism over U, see Categories, Lemma 0040. A morphism $(U, x, z, \alpha) \to (U', x', z', \alpha')$ is a pair of morphisms $a : x \to x'$ and $b : z \to z'$ compatible with α and α'. Thus it is clear that if f and g are faithful, so is the functor $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z} \to \mathcal{Y}$. Now, suppose that $(a, b), (a', b') : (U, x, z, \alpha) \to (U', x', z', \alpha')$ are two morphisms of the 2-fibre product. Then consider the equalizer $x'' \to x$ of a and a' and the equalizer $z'' \to z$ of b and b'. Since f commutes with equalizers (by Lemma lemma-fibre-products-morphism) we see that $f(x'') \to f(x)$ is the equalizer of $f(a)$ and

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$f(a')$. Similarly, $g(z'') \to g(z)$ is the equalizer of $g(b)$ and $g(b')$. Picture

$$\begin{array}{ccccc} f(x'') & \longrightarrow & f(x) & \overset{f(a)}{\underset{f(a')}{\rightrightarrows}} & f(x') \\ \downarrow{\scriptstyle \alpha''} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha'} \\ g(z'') & \longrightarrow & g(z) & \overset{g(b)}{\underset{g(b')}{\rightrightarrows}} & g(z') \end{array}$$

It is clear that the dotted arrow exists and is an isomorphism. However, it is not a priori the case that the image of α'' in the category of schemes is the identity of its source. On the other hand, the existence of α'' means that we can assume that x'' and z'' are defined over the same scheme and that the morphisms $x'' \to x$ and $z'' \to z$ have the same image in the category of schemes. Redoing the diagram above we see that the dotted arrow now does project to an identity morphism and we win. Some details omitted. $\square$